

Resource-Normalized Scaling Laws for Byzantine Consensus: Chance-Constrained Validator-Set Sizing from Commodity-CI Measurements

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Problem. Sizing a permissioned BFT validator set for e-governance must reconcile falling consensus throughput with rising fault-detection power as n grows. Published single-factor exponents are hard to interpret as protocol properties unless the concurrency path is reported. **Contributions.** A reproducible engineering methodology for resource-normalized calibration and capacity planning: *method* — resource-normalized measurement separating consensus cost from host contention; *analysis* — identifiability relation $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ under a bi-factor model; *engineering* — chance-constrained sizing with feasible set $[n_{\min}, n_{\max}]$; *empirics* — coverage-checked prediction intervals on public commodity CI. **Scope.** Four-core CI ($n \leq 16$), emulated RTT, and omission/duplication injection; fitted coefficients are deployment-specific and intended for recalibration under matching conditions.

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1. Introduction

Permissioned Byzantine Fault Tolerant (BFT) consensus underpins e-governance systems — voting, registries, digital identity — and raises a concrete procurement question: *how many validators should the network have?*

The validator count n pulls in opposite directions. Throughput argues for small n : every validator executes every transaction, and PBFT-family protocols exchange $O(n^2)$ messages per block. Detection reliability argues for large n . This paper develops a reproducible engineering methodology that combines the two under resource-normalized calibration.

A practical obstacle is that dedicated multi-host testbeds are often unavailable at design time. Commodity CI runners are accessible and reproducible, yet pack many validators onto few cores. Without normalization, fitted exponents mix consensus cost with host contention and with the load generator's concurrency schedule. Published single-factor fits $\text{TPS}(n) \propto n^\alpha$ then disagree in magnitude and sometimes in sign. Our earlier studies, extrapolated to sixty validators, differ by roughly an order of magnitude — often read as protocol-specific behaviour. Within the investigated setting we show that it can arise from measurement design

alone. Without reporting the concurrency path, published exponents are difficult to interpret as intrinsic protocol properties.

Contributions. The central methodological contribution is resource-normalized measurement (RNM): fixed CPU quota q per validator and $\widetilde{\text{TPS}} = \text{TPS}/q$, calibrating consensus cost separately from host contention. Analytically, under the unsaturated bi-factor model, $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ along a concurrency path (Proposition 1) interprets measured single-factor exponents and shows why an RNM factorial design ($\lambda = 0$) matters; the relation follows from model assumptions, not an empirical universal. We also formulate chance-constrained sizing with feasible set $[n_{\min}, n_{\max}]$ (Proposition 2), and calibrate the workflow on public Besu/QBFT commodity CI with coverage checks and a path consistency check within that configuration (Sections 5–6). Scope: throughput and detection under the evaluated deployment; absolute TPS environment-specific; cross-client replication out of scope. Coefficients are deployment-specific; the methodology is for recalibration under matching conditions.

2. Related work

BFT protocols. Classical PBFT [1] tolerates $f < n/3$ at $O(n^2)$ messages [2, 3]. Later designs — HotStuff [4], Tendermint [5], Algorand [6], HoneyBadgerBFT [7] — trade latency and randomization; Vukolić [8] argues that no protocol dominates. These lines of work fix the validator set; they do not size it under joint throughput and detection requirements.

Benchmarks. Fabric [9, 10], BLOCKBENCH [11] and BFT-SMaRt [12] report absolute throughput at chosen configurations. They do not separate concurrency from validator count, nor state measurement conditions under which fitted exponents would be expected to transfer.

Scalability characterization and tools. Amdahl [13], Gustafson [14] and Gunther’s USL [15] fold concurrency and system size into one load variable — the confound that RNM is designed to remove. Concentration bounds [16, 18], Kish’s design effect [17], delta-method intervals [20], the residual bootstrap [19] and chance-constrained programming [21, 22] are standard instruments. The proposed methodology couples throughput and detection through opposing monotonicities in n .

Prior work. Remote-voting analyses [23, 24, 25] place endpoint and coercion threats out of scope. Our earlier calibration [27] and switching study [28] left open an identifiable consensus exponent and sizing with measured detector rates.

3. Resource-normalized measurement and the bi-factor law

Capacity planning benefits from a calibrated exponent that reflects consensus cost rather than host contention or load-generator growth. We introduce an isolation protocol, a bi-factor characterization, and an identifiability result under stated assumptions.

3.1. Resource-normalized measurement as an isolation tool

If n validators share a host of Q cores, each receives Q/n cores, so measured throughput mixes algorithmic cost with a shrinking compute budget. RNM is designed to isolate those effects under controlled quotas.

Definition 1 (Resource-normalized measurement). *A campaign follows the RNM protocol with quota q if every validator container receives the same CPU quota and memory limit independently of n , with aggregate reservation nq below host capacity. Normalized throughput is $\text{TPS} := \text{TPS}/q$.*

Assumption 1 (Separable compute budget). On the operating range of quotas in the campaign, $\text{TPS}(n, c, q) = q \cdot \Phi(n, c) \cdot (1 + r)$ with $|r| \leq Cq$ for a constant C independent of (n, c) there.

Assumption check. Under Assumption 1 and a power-law factor $\Phi \propto n^{-\beta}$, the OLS slope of $\log \text{TPS}$ on $\log n$ at fixed concurrency is independent of q : the relative remainder is absorbed by the intercept. Section 5 tests this invariance across $q \in \{0.20, 0.25, 0.33\}$. Within the evaluated deployment the calibrated model is reported at the reference quota $q = 0.25$; outside that regime the numerical coefficients must be re-fit.

Definition 2 (Bi-factor scaling characterization). *Under the RNM protocol,*

$$\text{TPS}(n, c) = T_0 \cdot g(c) \cdot n^{-\beta}, \quad g(c) = \frac{c^\gamma}{1 + (c/c^*)^\theta}, \quad (1)$$

with $T_0 > 0$, $\beta > 0$, $\gamma \in (0, 1]$, $\theta > \gamma$, $c^* > 0$. In the unsaturated regime $c \ll c^*$ this is log-linear: $\log \text{TPS} = \log T_0 + \gamma \log c - \beta \log n$.

The form separates concurrency from validator-count scaling, reduces to a power law for $c \ll c^*$, saturates at large c , and contains the classical single-factor fit as the case $\gamma = 0$ (or $\lambda = 0$ with $\alpha = -\beta$).

3.2. Identifiability

Proposition 1 is analytical under the unsaturated bi-factor model and stated path assumptions. Its importance is interpretive: it shows how to read a measured single-factor exponent under a stated concurrency path, rather than introducing a fundamentally new mathematical identity. Section 5 checks consistency within the evaluated deployment; that check does not extend the claim beyond the model assumptions.

Proposition 1 (Identifiability of the consensus exponent). *Let observations follow the unsaturated bi-factor law with additive noise of mean zero and finite variance, along a concurrency path $\log c_i = \lambda \log n_i + \mu + \eta_i$ ($\mathbb{E}[\eta_i | n_i] = 0$). If the empirical second moment of $x_i = \log n_i$ converges to a positive limit, the OLS slope $\hat{\alpha}_m$ of $\log \text{TPS}_i$ on x_i alone satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$. In particular $\hat{\alpha} > 0 \iff \lambda > \beta/\gamma$, and $\hat{\alpha} = -\beta \iff \lambda = 0$.*

Proof. Write $x_i = \log n_i$, $y_i = \log \text{TPS}_i$. The path reduces the unsaturated law to $y_i = a + \alpha x_i + \varepsilon_i$ with $\mathbb{E}[\varepsilon_i | x_i] = 0$, finite variance, and

$$\alpha = -\beta + \gamma\lambda. \quad (2)$$

Centred OLS gives $\hat{\alpha}_m = \alpha + N_m/D_m$ with $N_m = \sum (x_i - \bar{x})\varepsilon_i$ and $D_m = \sum (x_i - \bar{x})^2$. The second-moment assumption yields $D_m/m \rightarrow \text{Var}^*(x) > 0$; the summands of N_m are uncorrelated, mean zero, and of bounded second moment, so $N_m/m \rightarrow 0$ in probability. Continuous mapping: $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \alpha$. Signs follow from $\gamma > 0$. ■

Thus $\alpha = \underbrace{-\beta}_{\text{protocol}} + \underbrace{\gamma\lambda}_{\text{measurement}}$. Only a factorial design ($\lambda = 0$) recovers the consensus exponent under the stated model. For illustration, $\gamma = 0.8$, $\beta = 0.5$ and $\lambda = 1$ yield $\hat{\alpha} = +0.3$, and extrapolating from $n = 5$ to $n = 60$ differs by a factor ≈ 7 , matching the order of the discrepancy in [27].

Corollary 1 (Non-transferability of single-factor exponents). *Under the unsaturated bi-factor model, laboratories measuring along $\lambda_1 \neq \lambda_2$ obtain exponents differing by $\gamma(\lambda_1 - \lambda_2)$. Without reporting λ , published single-factor exponents should not be treated as transferable protocol properties across measurement designs.*

3.3. Implementation

On a factorial grid ($\lambda = 0$) we estimate $(\log T_0, \gamma, \beta)$ by OLS in log-space and (c^*, θ) by non-linear least squares on the full c -sweep (Section 5). The calibrated model then feeds prediction intervals and the sizing program of the next section within the range supported by the campaign.

4. Prediction intervals, detection, and the sizing problem

A calibrated consensus exponent is an input to capacity planning. This section assembles an engineering workflow: a throughput chance constraint and an optional detection constraint. Any detector yielding measured (p_d, p_f, ρ) plugs in; experiment X5 is one instance under the evaluated deployment.

4.1. Definitions and assumptions

Definition 3 (Fault class and detector). *A subset B of size k is faulty if each member fails a proposal round independently with probability $p_d > p_f$ (omission class; equivocation excluded). The detector flags validator i when its participation over w rounds falls below $\tau \in (p_f, p_d)$; detection succeeds when at least one member of B is flagged.*

Assumption 2 (Exchangeable correlation). Participation indicators in B are exchangeable with pairwise correlation $\rho \in [0, 1)$, so the mean carries design effect $1 + (k - 1)\rho$ in the sense of Kish [17].

4.2. Prediction intervals and detection bound

With regressor $\mathbf{z} = (1, \log c, -\log n)^\top$ and $\hat{\vartheta} = (\log T_0, \gamma, \beta)^\top$, the delta method [20] gives predictive variance $\hat{V} = \mathbf{z}^\top \widehat{\text{Cov}}(\hat{\vartheta}) \mathbf{z} + \hat{\sigma}^2$. An approximate $(1 - \varepsilon_p)$ prediction interval for $\text{TPS}(n, c)$ is then $\exp\{\log \widehat{\text{TPS}} \mp z_{1-\varepsilon_p/2} \hat{V}^{1/2}\}$; the one-sided lower endpoint $\underline{\text{TPS}}_{1-\varepsilon_p}$ uses $z_{1-\varepsilon_p}$. A residual bootstrap [19] supplies empirical quantiles when residual normality is doubtful. Coverage is checked on held-out validator counts within the calibrated range (Section 5).

With $k = \lceil \varphi n \rceil$, a design-effect heuristic yields the approximate miss probability

$$p_{\text{miss}}(n, w) \lesssim \exp\left\{-\frac{2wk(p_d - \tau)^2}{1 + (k - 1)\rho}\right\}, \quad (3)$$

which is nonincreasing in n and w . The cut $p_{\text{miss}} \leq \varepsilon_s$ gives

$$n_{\min} = \left\lceil \frac{1 + (k-1)\rho}{\varphi} \cdot \frac{\log(1/\varepsilon_s)}{2w(p_d - \tau)^2} \right\rceil. \quad (4)$$

Expression (3) is an engineering approximation adapting Hoeffding [16] via Kish [17] ($n_{\text{eff}} = kw/(1 + (k-1)\rho)$), not a new exact bound. With X5 AUC = 0.506, the evaluated deployment defaults to the throughput-bound constraint.

4.3. The sizing program

Problem 1 (Validator-set sizing). Given peak demand D_{peak} , levels $\varepsilon_p, \varepsilon_s$, window w , faulty fraction φ , production scale-up $\kappa > 0$ from RNM units to absolute throughput, and nondecreasing $\text{Cost}(n)$, minimize $\text{Cost}(n)$ subject to $\mathbb{P}[\kappa \cdot \widehat{\text{TPS}}(n, c^*) \geq D_{\text{peak}}] \geq 1 - \varepsilon_p$ and $p_{\text{miss}}(n, w) \leq \varepsilon_s$, where $c^* = \arg \max g$. When κ is free, take the smallest value that meets the throughput constraint at $n = 4$.

The chance constraint is equivalent, via the prediction interval above, to

$$h(n) := \log(\kappa T_0 g(c^*)/D_{\text{peak}}) - \beta \log n - z_{1-\varepsilon_p} \hat{V}^{1/2}(n, c^*) \geq 0. \quad (5)$$

On the model-based extrapolation range the fitted covariance confirms that $\hat{V}$ is nondecreasing in $\log n$ (Section 5), so with $\beta > 0$ the map h is strictly decreasing there; on the measured grid we evaluate h directly.

Proposition 2 (Feasibility window). *Whenever h is strictly decreasing and the approximate detection cut (4) is used, the feasible set is $\{n : n_{\min} \leq n \leq n_{\max}\}$ with $n_{\min}$ from (4) and $n_{\max} = \lfloor \sup\{n : h(n) \geq 0\} \rfloor$. If nonempty and Cost is nondecreasing then $n^* = n_{\min}$.*

Proof. Detection holds on $\{n \geq n_{\min}\}$ and throughput on $\{n \leq n_{\max}\}$, hence an interval; monotone cost selects the left endpoint. ■

Corollary 2 (Empty feasible set). *If $n_{\min} > n_{\max}$, no validator count satisfies both requirements under the stated constraints: the architecture must change, not merely the provisioning.*

The window is the intersection of two monotone constraints [21, 22]. Within the proposed methodology it converts a calibrated model into a reproducible sizing workflow for the evaluated deployment class.

5. Experimental campaign on commodity CI

5.1. Setup and design

The campaign runs in a public continuous-integration workflow on a four-core, 16 GB hosted runner. Constrained CI is the intended calibration setting: accessible, reproducible, and usable when dedicated clusters are unavailable. Each job provisions a Hyperledger Besu [26] QBFT network of n validators under the RNM protocol (Definition 1). A closed-loop generator drives c concurrent submitters and records steady-state committed throughput after warm-up. Faults are injected with kernel traffic control (omission and duplication; not equivocation).

X1/X3 calibrate (β, γ) on a factorial grid; **X2** checks the quota regime; **X3b** re-samples along concurrency paths; **X4** varies emulated RTT; **X5** reports detector rates; **X7/X8** check

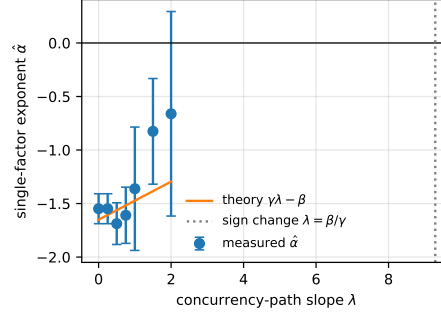


Fig. 1. Measured $\hat{\alpha}$ (points) vs. analytical $\gamma\lambda - \beta$ (line) in the evaluated deployment (Proposition 1). Sign change at $\lambda = \beta/\gamma = 9.29$

Table 1. Calibrated parameters, quota check, detector, and coverage (evaluated deployment)

Quantity	Estimate	Note
$T_0; \beta; \gamma$	2016.5; 1.651 ± 0.151 ; 0.178 ± 0.070	bi-factor
$c^*, \theta; \hat{\sigma}$	67.9, 2.542; 0.404	saturation / residual
$\hat{\alpha}$ at $\lambda = 0, 1, 2$	-1.548, -1.362, -0.661	X3b
β at $q = 0.20, 0.25, 0.33$	1.550, 1.541, 0.879	X2, $p = 0.002$
β at RTT 0/high	1.527, 2.708	X4
$p_d, p_f, \rho; \text{AUC}$	0.397, 0.376, 0.000; 0.506	X5, $\tau = 0.90$
coverage 90%/95%	90.9 %, 95.5 %	hold-out $n \in \{11, 13, 16\}$

prediction-interval coverage and sizing ablation. Cross-client replication (**X9**/GoQuorum) is out of scope. In total 94 successful records, ≈ 27.2 core-hours, $n_{\max} = 16$.

Reproducibility. All experiments are reproducible from the public CI workflow and archived artefacts at https://github.com/bpenyak/Resource_Normalized_Scaling_Laws_for_Byzantine_Consensus.

5.2. Results

X1/X3 – bi-factor calibration. Joint OLS on the reference-quota subset yields $\beta = 1.651 \pm 0.151$ and $\gamma = 0.178 \pm 0.070$ ($R^2 = 0.727$; Table 1). Under RNM one expects $\beta > 0$ when concurrency is held fixed. The fit confirms that expectation at $q = 0.25$ within the evaluated deployment.

X3b – path check of Proposition 1. A single-factor fit along $\lambda = 1$ gives $\hat{\alpha} = -1.362$ against $\gamma\lambda - \beta = -1.473$. Figure 1 shows measured exponents tracking $\gamma\lambda - \beta$ within the investigated configuration – consistency with the analytical prediction, not universal validity of the bi-factor form.

X2 – quota-regime check. The partial F -test for quota $\times \log n$ interaction rejects pooling across $\{0.20, 0.25, 0.33\}$ ($p = 0.002$): β drops from ≈ 1.541 at $q = 0.25$ to 0.879 at $q = 0.33$. Deployment-specific coefficients are therefore reported at the reference quota $q = 0.25$.

X4 – latency. Under elevated emulated RTT, β rises from 1.527 to 2.708 in this campaign. Calibration on LAN-like RTT is therefore optimistic for geo-distributed deployments unless re-calibrated.

X5 – detector. Omission/duplication detection yields $\text{AUC} = 0.506$ (Figure 2, near chance), so the workflow defaults to the throughput-bound constraint under the evaluated detector.

X7/X8 – coverage, comparison, and ablation. Hold-out coverage meets the nominal levels in Table 1 within the calibrated range. The fitted covariance confirms that $\hat{V}$ is nondecreasing

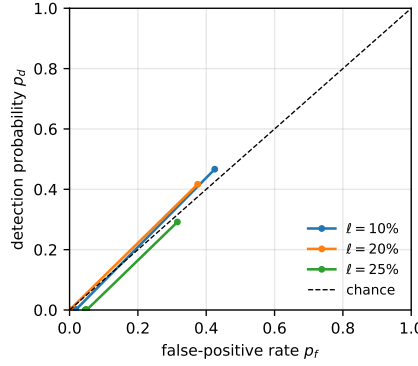


Fig. 2. Measured detector ROC (AUC = 0.506, near chance); throughput binds the window under the evaluated deployment

for $n \geq 16$, so h in (5) is strictly decreasing on the model-based extrapolation range; on the measured grid we evaluate h directly. Out-of-sample log-RMSE for linear / single-factor / USL [15] / bi-factor predictors is 1.329, 0.523, 0.637, 0.538. The bi-factor improvement over USL is consistent with decoupling concurrency from replica count under RNM. Relative sizing costs: 4.00, 1.50, 1.50, 1.00.

6. Application, implications, and limitations

6.1. Application and what-if analysis

We instantiate Problem 1 on $\approx 3 \times 10^7$ ballots over twelve hours ($D_{\text{peak}} \approx 2.1 \times 10^3$ TPS), requiring detection of a coordinated omission fault of fraction $\varphi = 0.2$ within one epoch at level $1 - \varepsilon_s$. Absolute demand enters via scale-up κ from RNM units. With free κ , Problem 1 takes

$$\kappa^* = \frac{D_{\text{peak}}}{\widetilde{\text{TPS}}_{1-\varepsilon_p}(4, c^*)}, \quad (6)$$

the minimal scale at which the lower prediction-interval bound at $n = 4$ meets D_{peak} .

Scenario A (baseline). Substituting the calibrated coefficients ($\beta = 1.651$, $\gamma = 0.178$) at $c^* = 32$ yields $\kappa = 13.00$ and feasibility window $[4, 4]$ with $n^* = 4$ (Proposition 2; Figure 3). Under this demand and hardware class the window collapses to a single admissible count: the network is throughput-bound at the smallest BFT set admitted by the formulation.

Scenario B (doubled peak demand, fixed hardware class). Hold $\kappa = 13.00$ and replace D_{peak} by $2D_{\text{peak}}$. Then $h(4) < 0$ and the feasible set is empty (Corollary 2): within the calibrated model, adding validators cannot restore doubled demand because predicted throughput falls with n .

Scenario C (WAN deployment, fixed hardware class). Retain $\kappa = 13.00$ and D_{peak} , but use $\beta = 2.708$ (X4) in place of $\beta = 1.527$. Again $h(4) < 0$: LAN-admissible hardware becomes infeasible under WAN β without raising κ .

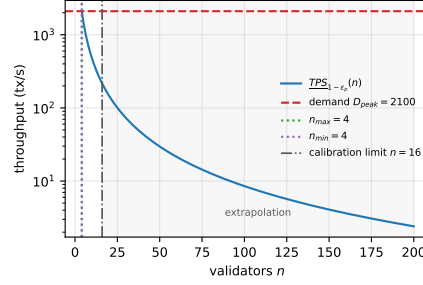


Fig. 3. Scenario A: calibrated lower PI throughput vs. n at $\kappa = 13.00$; window $[n_{\min}, n_{\max}] = [4, 4]$. Dash-dotted line: calibration limit $n = 16$ (measured region to the left; model-based extrapolation to the right)

The same workflow applies to registry and identity sizing after recalibration; binding elections on a blockchain remain outside the recommendation [23, 24, 25].

6.2. What this implies

A fitted (β, γ) should be treated as deployment-specific. Transfer is appropriate when concurrency path, quota regime, and RTT class match the calibration. The experiments indicate that future benchmarks should report λ (or $\lambda = 0$), per-validator quota, and RTT class (Corollary 1).

6.3. Limitations

Fault class. Encrypted transport limits X5 to omission/duplication. AUC 0.506 (near chance) motivates a throughput-bound window under the evaluated detector.

Infrastructure as design choice. Commodity four-core CI with $n \leq 16$ is the intended calibration substrate. Beyond measured counts, predictions are model-based; prediction intervals quantify extrapolation uncertainty.

Coefficients versus methodology. Coefficients (β, γ) are Besu/QBFT-specific and must not be copied across deployments. The RNM protocol and chance-constrained workflow transfer via recalibration under matching conditions; Proposition 1 remains analytical under its assumptions. Production scale-up uses κ in (6).

7. Conclusions

Analytical results. Under the unsaturated bi-factor model and stated path assumptions, the single-factor OLS exponent satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ (Proposition 1). Opposing monotone constraints yield a closed integer feasibility window (Proposition 2).

Experimentally supported observations. Under RNM on four-core commodity CI: $\beta = 1.651 \pm 0.151$, $\gamma = 0.178 \pm 0.070$; path re-sampling is consistent with $\gamma\lambda - \beta$ within the evaluated deployment; coefficients at $q = 0.25$; coverage meets nominal levels within the calibrated range; AUC = 0.506 defaults the window to throughput. WAN RTT raises β from 1.527 to 2.708; doubled demand or WAN β at fixed κ empties the window under the calibrated model. Numerical results are deployment-specific.

Engineering implications. The contribution is a reproducible methodology for resource-normalized calibration and capacity planning under RNM and calibrated chance constraints. Commodity CI is the intended calibration substrate; other deployments require recalibration of (β, γ) (and, where relevant, (p_d, p_f, ρ)) while reusing the same workflow and public artefacts.

Conflict of interest

The authors declare no conflict of interest.

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Data availability

Workflow definition, generators, fault-injection and analysis code, and raw artefacts are available at https://github.com/bpenyak/Resource_Normalized_Scaling_Laws_for_Byzantine_Consensus.

Use of artificial intelligence

AI-assisted tools were used for language editing and typesetting. All scientific results, proofs, design and analysis were produced by the authors.

Contributions of authors

Peniak B. O.: conceptualization, methodology, formal analysis, software, investigation, data curation, writing — original draft, visualization. **Liubinskyj B. B.:** supervision, validation, writing — review and editing.

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РЕСУРСНО-НОРМОВАНІ ЗАКОНИ МАСШТАБУВАННЯ BFT-КОНСЕНСУСУ: ІМОВІРНІСНО-ОБМЕЖЕНЕ ВИЗНАЧЕННЯ РОЗМІРУ МНОЖИНИ ВАЛІДАТОРІВ ЗА ВИМІРЮВАННЯМИ НА ТИПОВІЙ СІ-ІНФРАСТРУКТУРІ

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Відтворювана методика ресурсно-нормованого калібрування та планування потужності для BFT-валідаторів: пропускна здатність спадає зі зростанням n , а надійність виявлення відмов зростає з n . Внески: ресурсно-нормовані вимірювання; співвідношення ідентифікованості ($\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$); задача з імовірнісними обмеженнями та вікном $[n_{\min}, n_{\max}]$; прогнозні інтервали з перевіркою покриття на типовій СІ. Коефіцієнти специфічні для розгортання; процес розрахований на перекалібрування.

Ключові слова: візантійська відмовостійкість, консенсус, закон масштабування, планування потужності, оптимізація з імовірнісними обмеженнями, електронне урядування

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